The PV relationship can be understood as follows. The pressure is due to the force of the gas particles on the walls of the container. If a given amount of gas is compressed to half of its volume, the density is doubled and the number of particles hitting the unit area of the container will be doubled. Hence, the pressure would increase twofold.

## Consequence of Boyle's law

The pressure-density relationship can be derived from the Boyle's law as shown below.

$$P_1V_1 = P_2V_2 \quad \text{(Boyle's law)}$$

$$P_1 \frac{m}{d_1} = P_2 \frac{m}{d_2}$$

where "m" is the mass, $d_1$ and $d_2$ are the densities of gases at pressure $P_1$ and $P_2$.

$$\frac{P_1}{d_1} = \frac{P_2}{d_2} \quad \text{------ (6.4)}$$

In other words, the density of a gas is directly proportional to pressure.




**All the passenger aeroplane cabins have to be artificially pressurised. do you know why?**

Most commercial aeroplanes fly at about 30,000 feet altitude. The pressure decreases with the increase in altitude as there are fewer molecules per unit volume of air. Hence, while at air, the pressure around the aeroplane will be so low that one could pass out for lack of oxygen. For this reason aeroplanes cabins are artificially pressurized.



Similarly, the effect of drop in pressure is felt as a little pain in the ears by a person while ascending a mountain in a plain. Though the external pressure drops, the internal pressure within the ear cavities remains the same. This creates an imbalance. The greater internal pressure forces the eardrum to bulge outward causing pain. The excess air within the ear cavities escapes after some time and with the help of yawning and thereby equalizing the internal and external pressure to relieve the pain.

***Underwater divers are advised not to hold the breath unnecessarily while diving. Do you know why?***

In figure (6.3) let us find the missing parameters (volume in 6.3 (b) and pressure in 6.3(c))

Fig. 6.3(a) Fig. 6.3 (b) Fig. 6.3(c)



| | | |
|---|---|---|
| $P_1$ = 1 atm. | $P_2$=2 atm. | $P_3$=? atm. |
| $V_1$ = 1 $dm^3$ | $V_2$ = ? $dm^3$ | $V_3$ = 0.25$dm^3$ |
| T = 298 K | T = 298 K | T = 298 K |

**Figure. 6.3 Effect of pressure on volume of the gas to verify Boyle's law**



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